Hybrid Simulation

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1 Introduction

The wavefield injection/extrapolation can be viewed as solving a source-free interface discontinuity problem:

$$\begin{aligned} &\rho\frac{\partial^2 \boldsymbol{w}}{\partial t^2} - \nabla\cdot(\boldsymbol{C}:\nabla\boldsymbol{w}) = 0 \quad \text{in } \oplus \\ &\boldsymbol{w}\Big|_{\Gamma+}^{\Gamma-} = \boldsymbol{u}^0\Big|_{\Gamma} \\ &\hat{\boldsymbol{n}}\cdot(\boldsymbol{C}:\nabla\boldsymbol{w})\Big|_{\Gamma+}^{\Gamma-} = \hat{\boldsymbol{n}}\cdot(\boldsymbol{C}:\nabla\boldsymbol{u}^0)\Big|_{\Gamma} \end{aligned} \tag{1}$$

where $\oplus$ is the whole domain (including external source and the study region), Ω is the study region, and Γ is the boundary used for injection. $\boldsymbol{u}^0$ is the background field, and $\boldsymbol{w}$ is an auxiliary field defined as:

$$\boldsymbol{w} := \begin{cases} \boldsymbol{u} & \text{in } \Omega, \\ \boldsymbol{u} - \boldsymbol{u}^0 & \text{in } \oplus\backslash\bar{\Omega}. \end{cases} \tag{2}$$

To solve this equation, we usually have two ways: the direct simulation by incorporating the interface discontinuity, or by using representation theorem.

2 Non-hangling Node SEM Implementation of the Interface Discontinuity Problem

In this section, we derive an SEM scheme for the interface discontinuity problem, as defined by eq. (1). To obtain the weak form, we multiply both sides of eq. (1) by a test function ϕ, and integrate over $\oplus$. Using integration by parts, we obtain

$$\int_{\oplus}\rho\phi\frac{\partial^2 \boldsymbol{w}}{\partial t^2}\mathrm{d}V + \int_{\oplus}\nabla\phi\cdot\boldsymbol{C}:\nabla\boldsymbol{w}\mathrm{d}V = \int_{\Gamma}\phi\Delta\boldsymbol{T}\mathrm{d}S. \tag{3}$$

Note that here we require ϕ to be continuous across Γ. To solve the interface discontinuity problem, we need to find $\boldsymbol{w}$ that satisfies eq. (3) for any admissible test function ϕ. Based on SEM, the computational domain can be first discretized into elements with Gauss-Legendre-Lobatto (GLL) points, and the wavefield $\boldsymbol{w}$ is approximated on these GLL points.

We suppose that the space is discretized in a way that the interface is *honored*, i.e., the interface is defined only by points on element boundaries and does not run through any element. We use w_Γ^- and w_Γ^+ to denote the displacements for points on the negative and positive (the normal vector is pointing from negative to positive) side of Γ, and w_i for points not on Γ, then the discretized version of eq. (3) can be written as

$$\begin{bmatrix} \mathsf{M}_{\Gamma\Gamma}^- & \mathsf{M}_{\Gamma\Gamma}^+ & 0 \\ 0 & 0 & \mathsf{M}_{ii} \end{bmatrix} \begin{bmatrix} \ddot{\mathsf{w}}_\Gamma^- \\ \ddot{\mathsf{w}}_\Gamma^+ \\ \ddot{\mathsf{w}}_i \end{bmatrix} + \begin{bmatrix} \mathsf{K}_{\Gamma\Gamma}^- & \mathsf{K}_{\Gamma\Gamma}^+ & \mathsf{K}_{\Gamma i} \\ \mathsf{K}_{i\Gamma}^- & \mathsf{K}_{i\Gamma}^+ & \mathsf{K}_{ii} \end{bmatrix} \begin{bmatrix} \mathsf{w}_\Gamma^- \\ \mathsf{w}_\Gamma^+ \\ \mathsf{w}_i \end{bmatrix} = \begin{bmatrix} \Delta\mathsf{T}_\Gamma \\ 0 \end{bmatrix}, \tag{4}$$

in which $\mathsf{M}_{\Gamma\Gamma}^-$, $\mathsf{M}_{\Gamma\Gamma}^+$ and M_{ii} are positive-definite diagonal matrices in the SEM formulation. The displacement discontinuity can be written as

$$\mathsf{w}_\Gamma^- - \mathsf{w}_\Gamma^+ = \Delta\mathsf{w}_\Gamma. \tag{5}$$

Eqs. (4) and (5) together form a solvable system of ordinary differential equations. Directly solving this system requires treating each point on the interface as two separated GLL points, representing the positive and negative sides of the interface, respectively. This leads to a split-node scheme. To avoid this, we use the jump condition to eliminate w_Γ^+ from eq. (4), resulting in

$$\begin{bmatrix} \mathsf{M}_{\Gamma\Gamma}^- + \mathsf{M}_{\Gamma\Gamma}^+ & 0 \\ 0 & \mathsf{M}_{ii} \end{bmatrix} \begin{bmatrix} \ddot{\mathsf{w}}_\Gamma^- \\ \ddot{\mathsf{w}}_i \end{bmatrix} + \begin{bmatrix} \mathsf{K}_{\Gamma\Gamma}^- & \mathsf{K}_{\Gamma\Gamma}^+ & \mathsf{K}_{\Gamma i} \\ \mathsf{K}_{i\Gamma}^- & \mathsf{K}_{i\Gamma}^+ & \mathsf{K}_{ii} \end{bmatrix} \begin{bmatrix} \mathsf{w}_\Gamma^- \\ \mathsf{w}_\Gamma^- - \Delta\mathsf{w}_\Gamma \\ \mathsf{w}_i \end{bmatrix} = \begin{bmatrix} \Delta\mathsf{T}_\Gamma + \mathsf{M}_{\Gamma\Gamma}^+ \Delta\ddot{\mathsf{w}}_\Gamma \\ 0 \end{bmatrix}. \tag{6}$$

This scheme does not require the split-node implementation and only keeps the wavefield on the negative side of the interface w_Γ^-, which is easy to implement in SEM. Comparing with a typical wave simulation, only two additional steps are needed to incorporate interface discontinuity: (1) before applying the stiffness matrix in the elements that are on the negative side of the interface, an additional step is needed to add the displacement discontinuity $\Delta\mathsf{w}_\Gamma$ for the points on the interface, and (2) an additional force term $\Delta\mathsf{T}_\Gamma + \mathsf{M}_{\Gamma\Gamma}^+\Delta\ddot{\mathsf{w}}_\Gamma$ needs to be included for points on the interface.

In principle, we can eliminate either w_Γ^- or w_Γ^+ from eq. (6). However, when using this algorithm to simulate wavefield injection, it is often desirable to implement an PML-type absorbing boundary, immediately outside the local domain to absorb the scattered wavefield. Eliminating w_Γ^- and keeping w_Γ^+ allows fully independent implementation of absorbing boundary from wavefield injection. So if we want to keep $\boldsymbol{w}^-$, we should let the injection boundary not coincide with the PML boundary.

3 Equivalent Forces and Moment Tensors

The solution to Eq. (1) can alternatively be expressed using the representation theorem:

$$\begin{aligned} u_i(\boldsymbol{x},\omega) = &\int_\Sigma \mathrm{d}\Sigma\ \left[u_k(\boldsymbol{\xi},\omega)\right]_-^+ n_j c_{kjpq} \frac{\partial G_{ip}(\boldsymbol{x},\boldsymbol{\xi},\omega)}{\partial \xi_q} \\ &- \int_\Sigma \mathrm{d}\Sigma\ G_{ip}(\boldsymbol{x},\boldsymbol{\xi},\omega)\left[T_p(\boldsymbol{\xi},\omega)\right]_-^+ . \end{aligned} \tag{7}$$

By introducing a Dirac delta function, we can transform the 2-D integration variable $\boldsymbol{\xi}$ into the 3-D variable $\boldsymbol{\eta}$ via the following identities:

$$\begin{aligned} G_{ip}(\boldsymbol{x},\boldsymbol{\xi},\omega) &= \int_V \mathrm{d}V(\boldsymbol{\eta})\ G_{ip}(\boldsymbol{x},\boldsymbol{\eta},\omega)\delta(\boldsymbol{\eta}-\boldsymbol{\xi}), \\ \frac{\partial}{\partial \xi_q} G_{ip}(\boldsymbol{x},\boldsymbol{\xi},\omega) &= -\int_V \mathrm{d}V(\boldsymbol{\eta})\ G_{ip}(\boldsymbol{x},\boldsymbol{\eta},\omega)\frac{\partial}{\partial \eta_q}\delta(\boldsymbol{\eta}-\boldsymbol{\xi}). \end{aligned} \tag{8}$$

Substituting these relations into Eq. (7), the displacement field can be expressed as a volume integral:

$$\begin{aligned} u_i(\boldsymbol{x},\omega) = &\int_V \mathrm{d}V(\boldsymbol{\eta})\ G_{ip}(\boldsymbol{x},\boldsymbol{\eta},\omega) f_p^u(\boldsymbol{\eta},\omega) \\ &+ \int_V \mathrm{d}V(\boldsymbol{\eta})\ G_{ip}(\boldsymbol{x},\boldsymbol{\eta},\omega) f_p^T(\boldsymbol{\eta},\omega), \end{aligned} \tag{9}$$

where the equivalent force terms are defined as:

$$\begin{aligned} f_p^T(\boldsymbol{\eta},\omega) &= -\int_\Sigma \mathrm{d}\Sigma\ \left[T_p(\boldsymbol{\xi},\omega)\right]_-^+ \delta(\boldsymbol{\xi}-\boldsymbol{\eta}), \\ f_p^u(\boldsymbol{\eta},\omega) &= \int_\Sigma \mathrm{d}\Sigma\ \left[u_k(\boldsymbol{\xi})\right]_-^+ n_j c_{kjpq} \frac{\partial}{\partial_{\xi_q}}\delta(\boldsymbol{\xi}-\boldsymbol{\eta}). \end{aligned} \tag{10}$$

Equation (9) corresponds to the simultaneous simulation of distributed force sources along the injection boundary. In the context of the spectral-element method (SEM), the injection interface is constrained to lie along element boundaries; therefore, these surface integrals can be evaluated using Gauss-Lobatto-Legendre (GLL) quadrature. It remains to express the δ function through an expansion of Lagrange polynomials:

$$\delta(\boldsymbol{x}-\boldsymbol{x}^s) = \sum_{i,j=1}^{N_{gll}} a_{ij} l_i(\xi) l_j(\eta). \tag{11}$$

For an arbitrary function $f(\boldsymbol{x})$, the sifting property implies:

$$\int \mathrm{d}\Sigma\ f(\boldsymbol{x})\delta(\boldsymbol{x}-\boldsymbol{x}^s) = f(\boldsymbol{x}^s) = \sum_{i,j=1}^{N_{gll}} f_{ij} l_i(\xi^s) l_j(\eta^s), \tag{12}$$

yielding the coefficients:

$$a_{ij} = \frac{l_i(\xi^s) l_j(\eta^s)}{w_i w_j J^{ij}}, \tag{13}$$

where J^{ij} is the 2-D Jacobian at a given GLL point. If the source location $\boldsymbol{x}^s$ coincides exactly with a GLL point (ξ_k, η_l), this term simplifies to:

$$a_{ij} = \frac{\delta_{ik}\delta_{jl}}{\omega_k \omega_l J^{kl}}. \tag{14}$$

Consequently, the equivalent forces f_p^T and f_p^u at a specific GLL point are evaluated as:

$$\begin{aligned} f_p^T(\xi_m, \eta_n) =& - \left[T_p^{mn}\right]_-^+, \\ f_p^u(\xi_m, \eta_n) =& \frac{1}{\omega_m J^{mn}} \sum_a M_{pq}^{an} J^{an} \omega_a l'_m(\xi^a) \left[\frac{\partial \xi}{\partial x_q}\right]^{an} \\ & + \frac{1}{\omega_n J^{mn}} \sum_b M_{pq}^{mb} J^{mb} \omega_b l'_n(\eta^b) \left[\frac{\partial \eta}{\partial x_q}\right]^{mb}, \end{aligned} \tag{15}$$

where M_{pq} is defined as:

$$M_{pq} = [u_k]_-^+ c_{kjpq} n_j. \tag{16}$$

Alternatively, we can interpret this term as an equivalent moment tensor density:

$$M_{pq}(\xi_m, \eta_n) = [u_k]_-^+ c_{kjpq} n_j. \tag{17}$$

Thus, the injection problem is reduced to the simulation of a distribution of point forces and moment tensors along the injection boundary.